

Section: Electricity

Topic: Semiconductors, Sources, Internal Resistance

This question involves the behaviour of an LED and the I-V characteristics of a circuit component.

(a)

State the minimum potential difference across the LED required for it to conduct

✔ From the I–V graph shown, the LED begins to conduct at:
2.0 V

(b)

Use the graph to determine the resistance of the component at a potential difference of 4.0 V

From the graph:

- At $V = 4.0\text{ V}$, $I = 0.050\text{ A}$

Use Ohm’s Law:

$$R = \frac{V}{I} = \frac{4.0}{0.050} = 80\ \Omega$$

(c)

Explain why the LED does not conduct when the potential difference is less than 2.0 V

✔ The LED has a **threshold voltage** (here 2.0 V).
When the potential difference is **below this value**, the **energy is insufficient to overcome the potential barrier** at the p–n junction, so **no current flows**.

🧠 Revision Tips

- LEDs have a **minimum voltage** (threshold) before they conduct — usually between 1.8 V and 3.3 V
- Always use **I–V graph data points** directly to calculate resistance
- Resistance is **non-linear** in components like diodes and LEDs — not constant
- Below threshold voltage, the LED behaves like an **open switch**